Supplementary Information

Structural bioinformatic and evolutionary studies of synaptic vesicle glycoprotein 2 family members, synaptophysin, synaptogyrins, and their AlphaFold3 predicted water-soluble QTY variants and uncovering the T<=>V co-evolution

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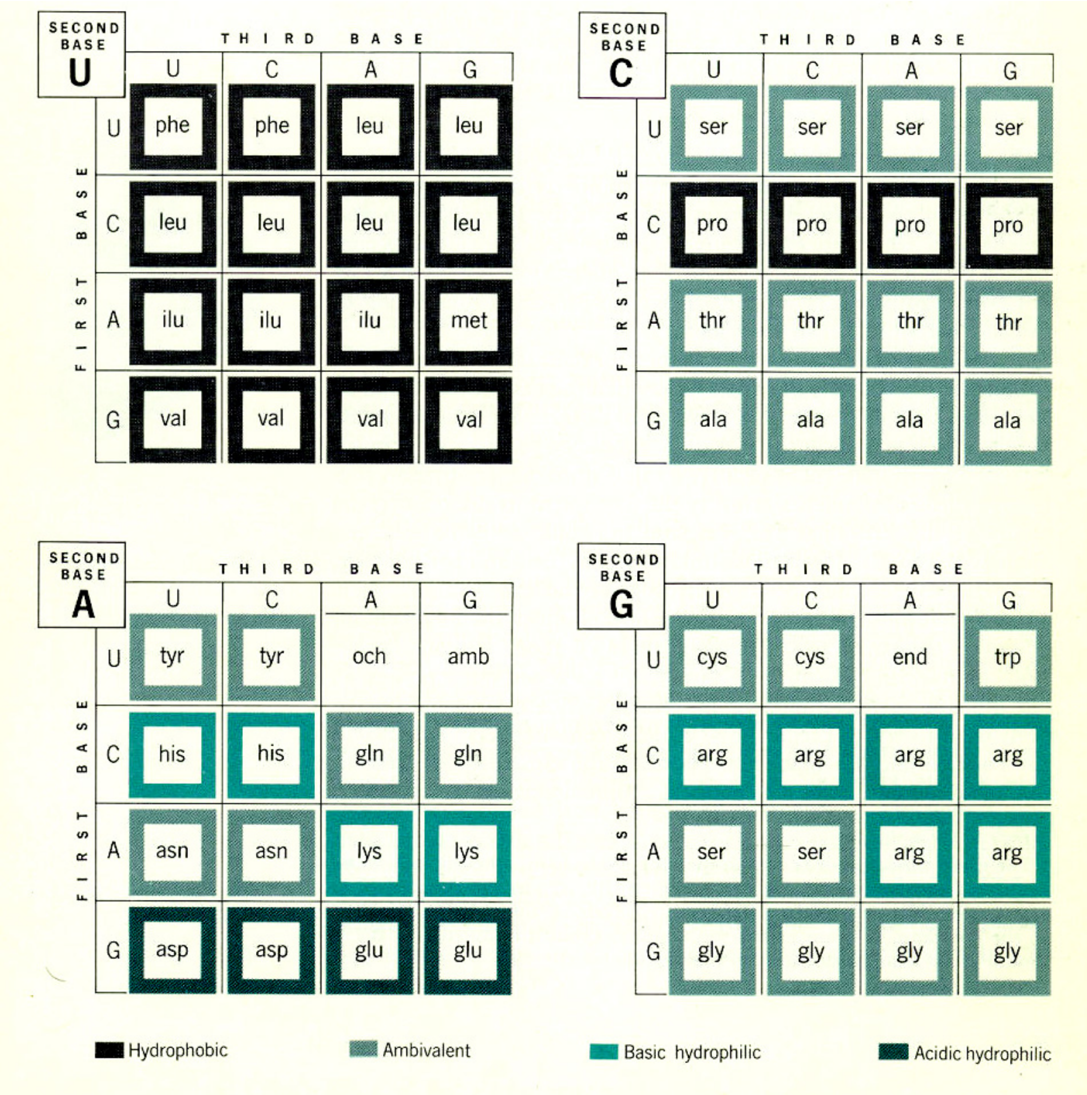


Figure S1. The chemical nature of amino acids is primarily determined by the second position of their genetic code. The second position is particularly significant, as seen in several examples:

- **i.** Amino acids with uracil (U) in the second position tend to be hydrophobic, including phenylalanine (Phe), leucine (Leu), isoleucine (Ile), valine (Val), and methionine (Met).
- **ii.** Amino acids with cytosine (C) in the second position are either less hydrophobic, such as proline (Pro) and alanine (Ala), or contain a hydroxyl (-OH) group, like serine (Ser) and threonine (Thr).
- **iii.** Amino acids with adenine (A) in the second position are generally hydrophilic and water-soluble. These include aspartic acid (Asp), glutamic acid (Glu), asparagine (Asn), lysine (Lys), histidine (His), and tyrosine (Tyr). Additionally, two stop codons, ochre (UAA) and amber (UAG), have A in the second position.
- **iv.** Amino acids with guanine (G) in the second position are also water-soluble. Examples are arginine (Arg) and serine (Ser). Cysteine (Cys) is partially water-soluble, and glycine (Gly) is achiral with a hydrogen atom as its side chain. The stop codon UGA is also noted here.

Overall, the presence of pyrimidines (U and C) at the second position is associated with hydrophobic properties, whereas purines (A and G) contribute to hydrophilicity.

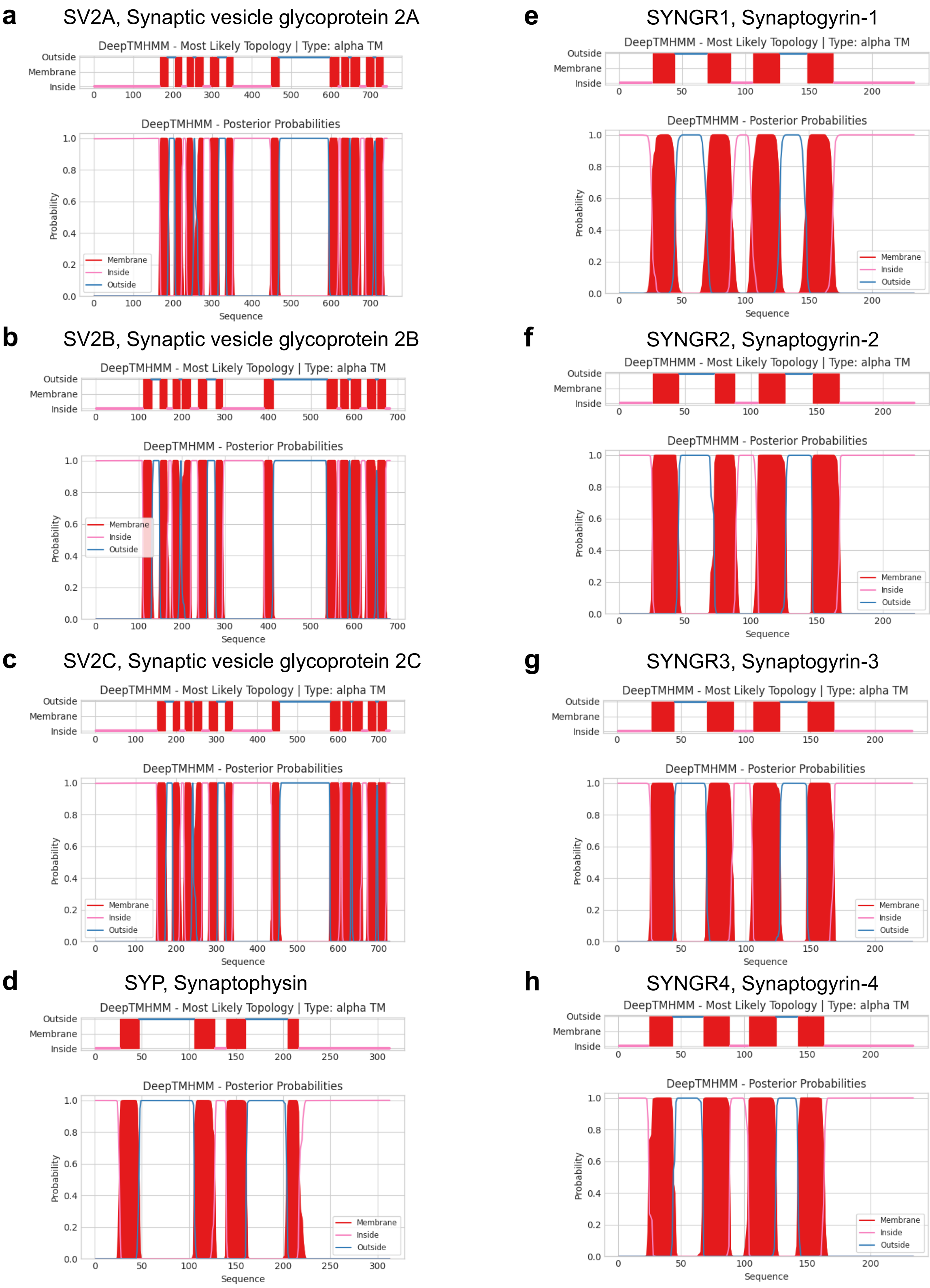


Figure S2. Membrane topology models of eight native synaptic vesicle proteins. Transmembrane (TM) helix predictions were carried out using DeepTMHMM, a deep learning model for transmembrane topology prediction and classification.

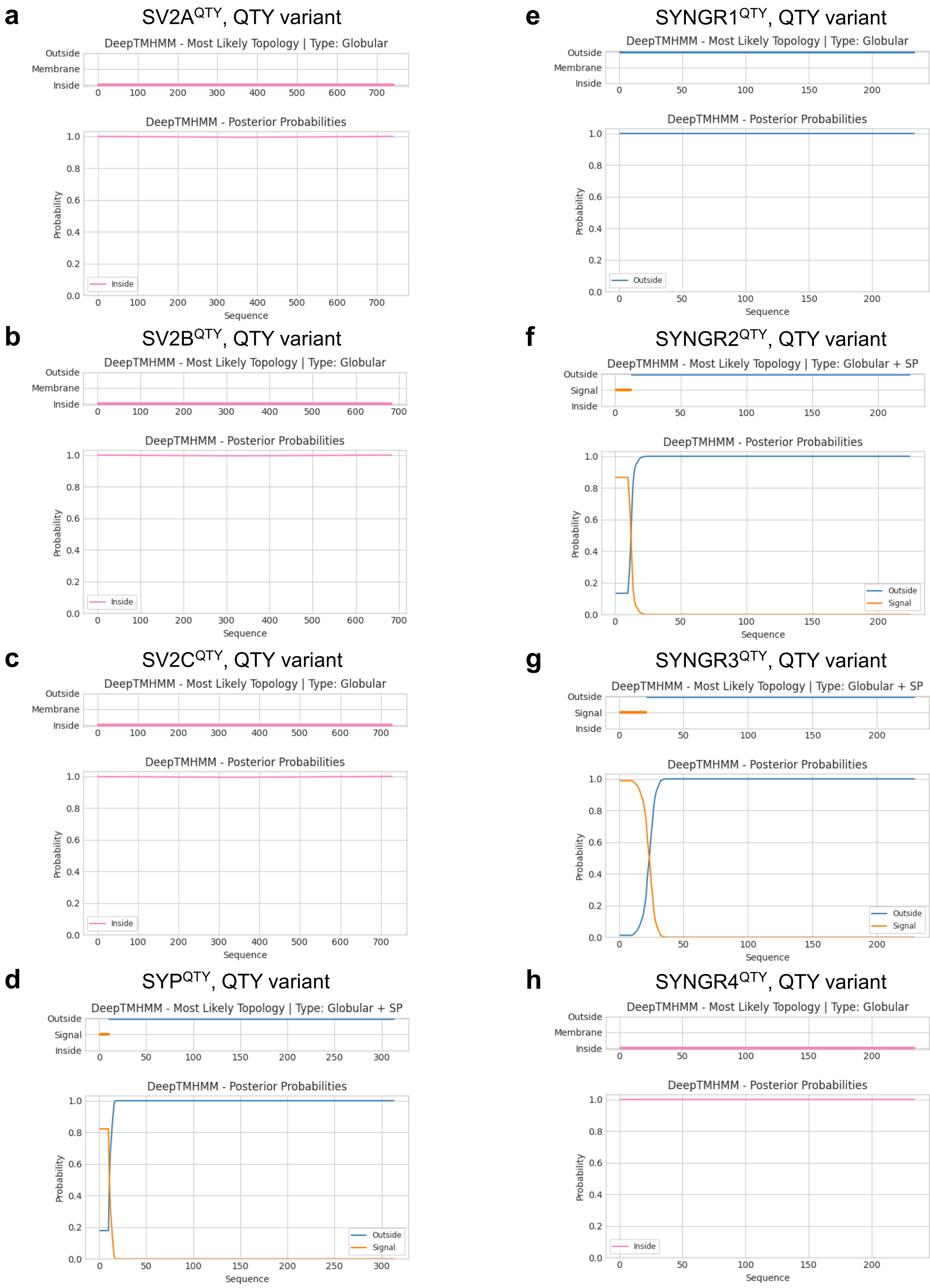


Figure S3. Membrane topology models of QTY variants of eight synaptic vesicle proteins. Transmembrane (TM) helix predictions were carried out using DeepTMHMM, a deep learning model for transmembrane topology prediction and classification.

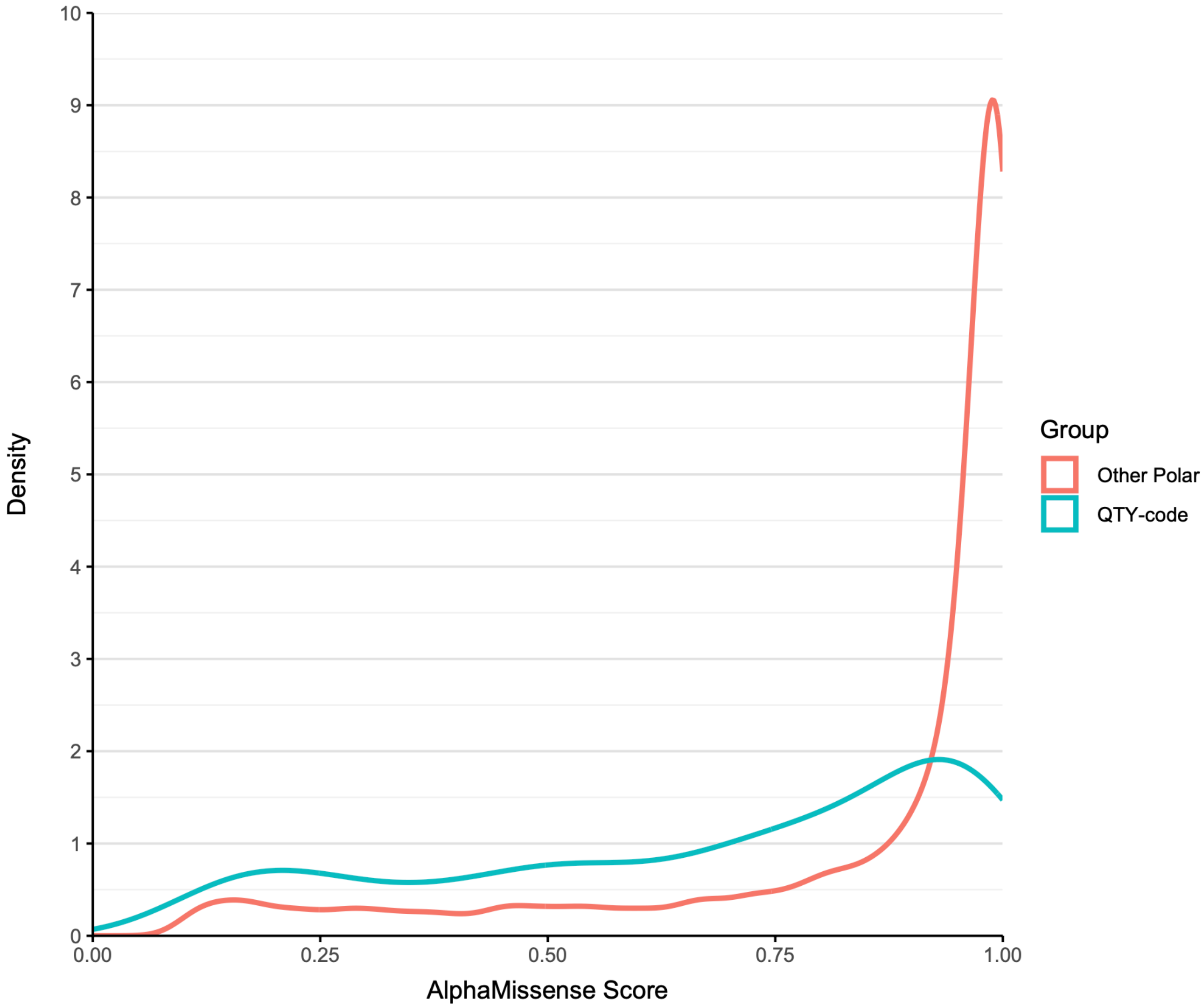


Figure S4. Density Plot of predicted effects of variations at the residues L, I, F of Synaptic vesicle glycoproteins A-C (SV2A-C). The x-axis indicates the scores of the AlphaMissense predicted effect of the substitution, ranging from benign (0.0) to damaging (1.0). The y-axis indicates the density of the score. Median score for QTY variants was M=0.752 (0.470-0.926), median score for other polar substitutions was M=0.972 (814-0.994). Kolmogorov Smirnov test indicated a significant difference between the two distributions (D(477,4288) = 0.4, p =<0.001).

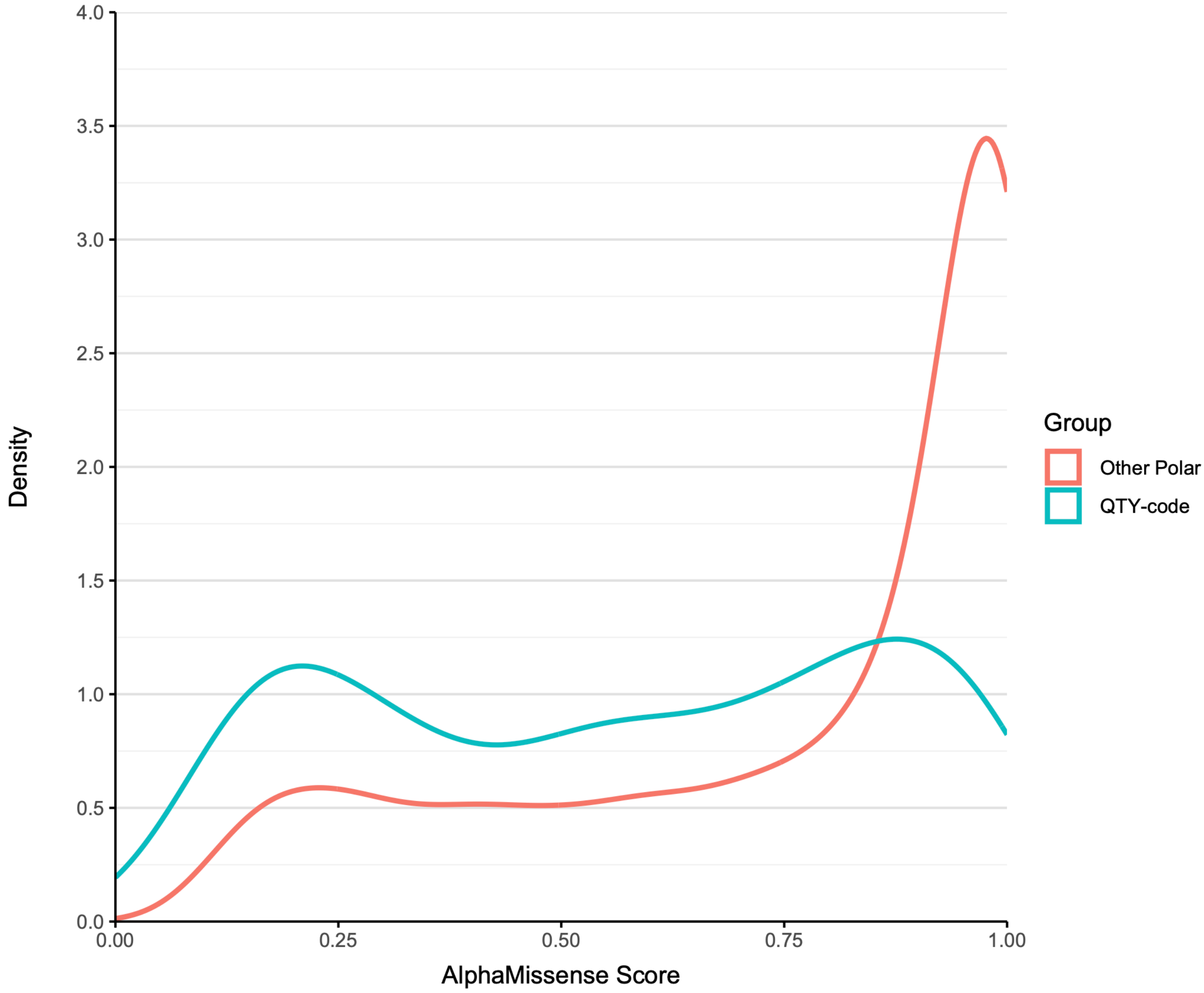


Figure S5. Density Plot of predicted effects of variations at the residues L, I, F of Synaptogyrins (SYNGR1-4) and Synaptophysin (SYP). The x-axis indicates the scores of the AlphaMissense predicted effect of the substitution, ranging from benign (0.0) to damaging (1.0). The y-axis indicates the density of the score. Median score for QTY variants was M= 0,580 (0,286-0,833), median score for other polar substitutions was M=0,899 (0,570-0,989). Kolmogorov Smirnov test indicated a significant difference between the two distributions (D(248,2229) = 0.34, p <0.001).

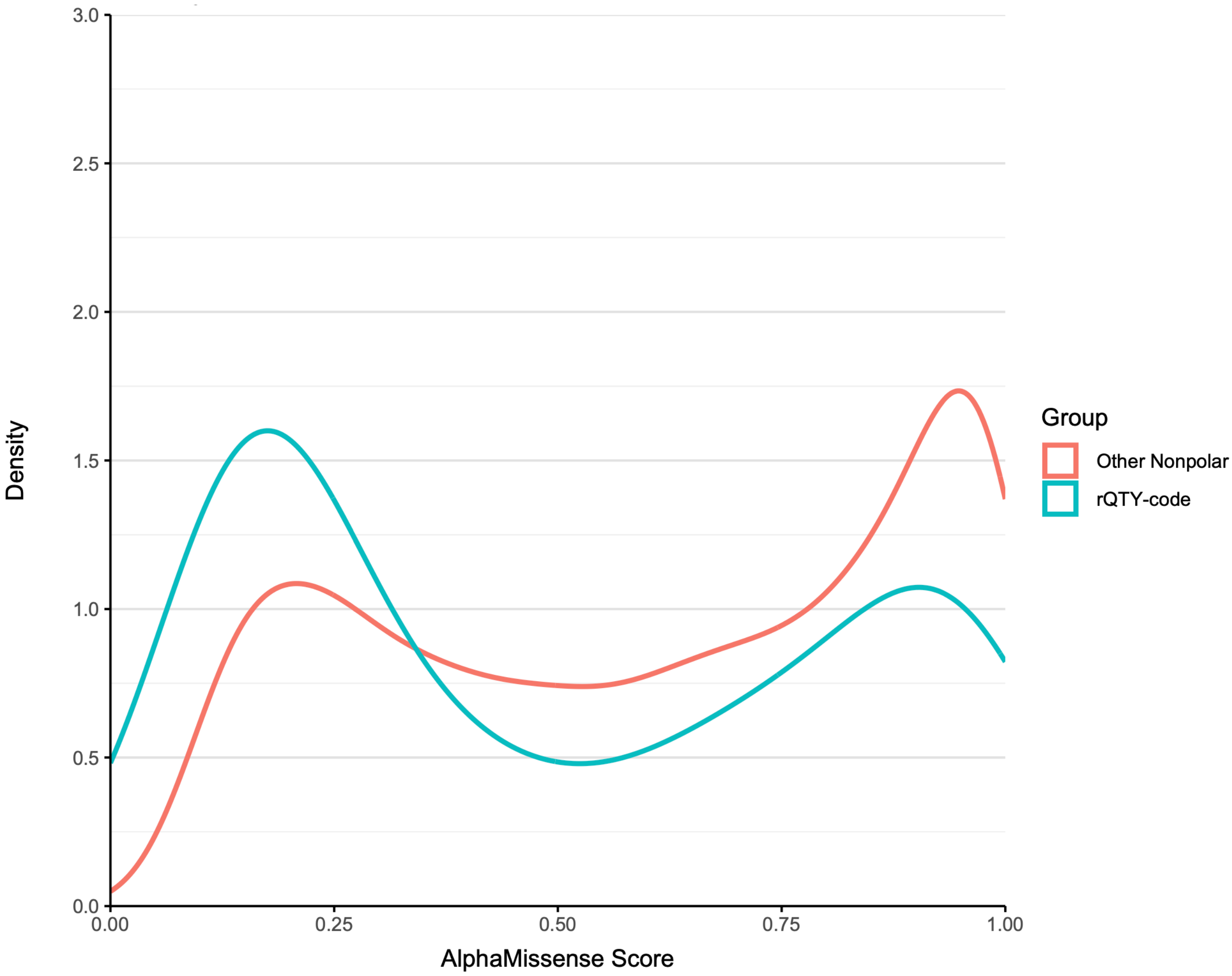


Figure S6. Density Plot of predicted effects of variations at the residues Q, T, Y of Synaptic vesicle glycoproteins A-C (SV2A-C). The x-axis indicates the scores of the AlphaMissense predicted effect of the substitution, ranging from benign (0.0) to damaging (1.0). The y-axis indicates the density of the score. Median score for rQTY variants was M=0.399 (0.175-0.842) median score for other nonpolar substitutions was M=0.652 (0.324-0.895). Kolmogorov Smirnov test indicated a significant difference between the two distributions (D(2337,260) = 0.22, p <0.001).

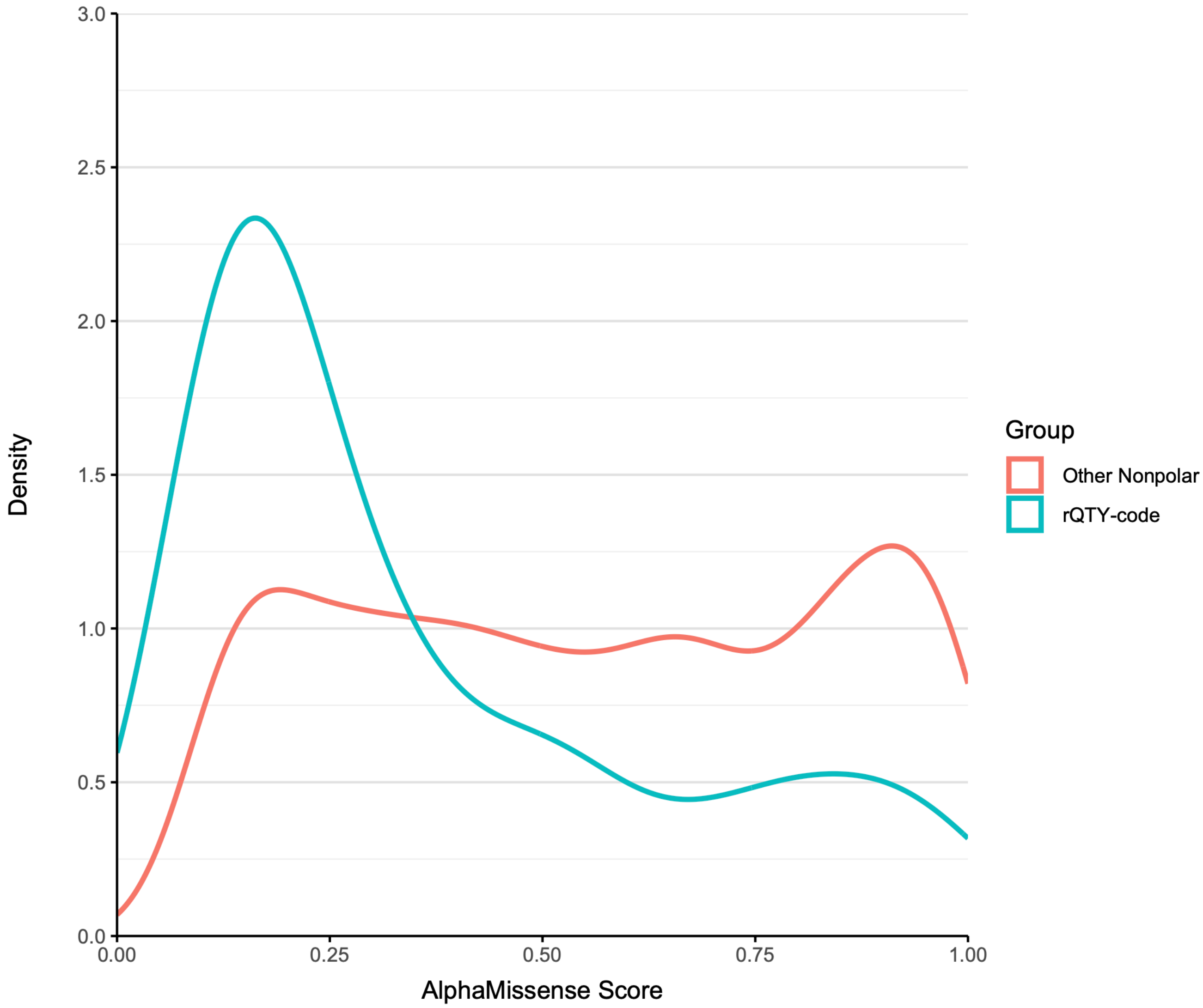


Figure S7. Density Plot of predicted effects of variations at the residues Q, T, Y of Synaptogyrins (SYNGR1-4) and Synaptophysin (SYP). The x-axis indicates the scores of the AlphaMissense predicted effect of the substitution, ranging from benign (0.0) to damaging (1.0). The y-axis indicates the density of the score. Median score for rQTY variants was M= 0.240 (0.141-0.519) median score for other nonpolar substitutions was M= 0.8993 (0.570-0.989). Kolmogorov Smirnov test indicated a significant difference between the two distributions (D(1655,184) = 0.35, p <0.001)

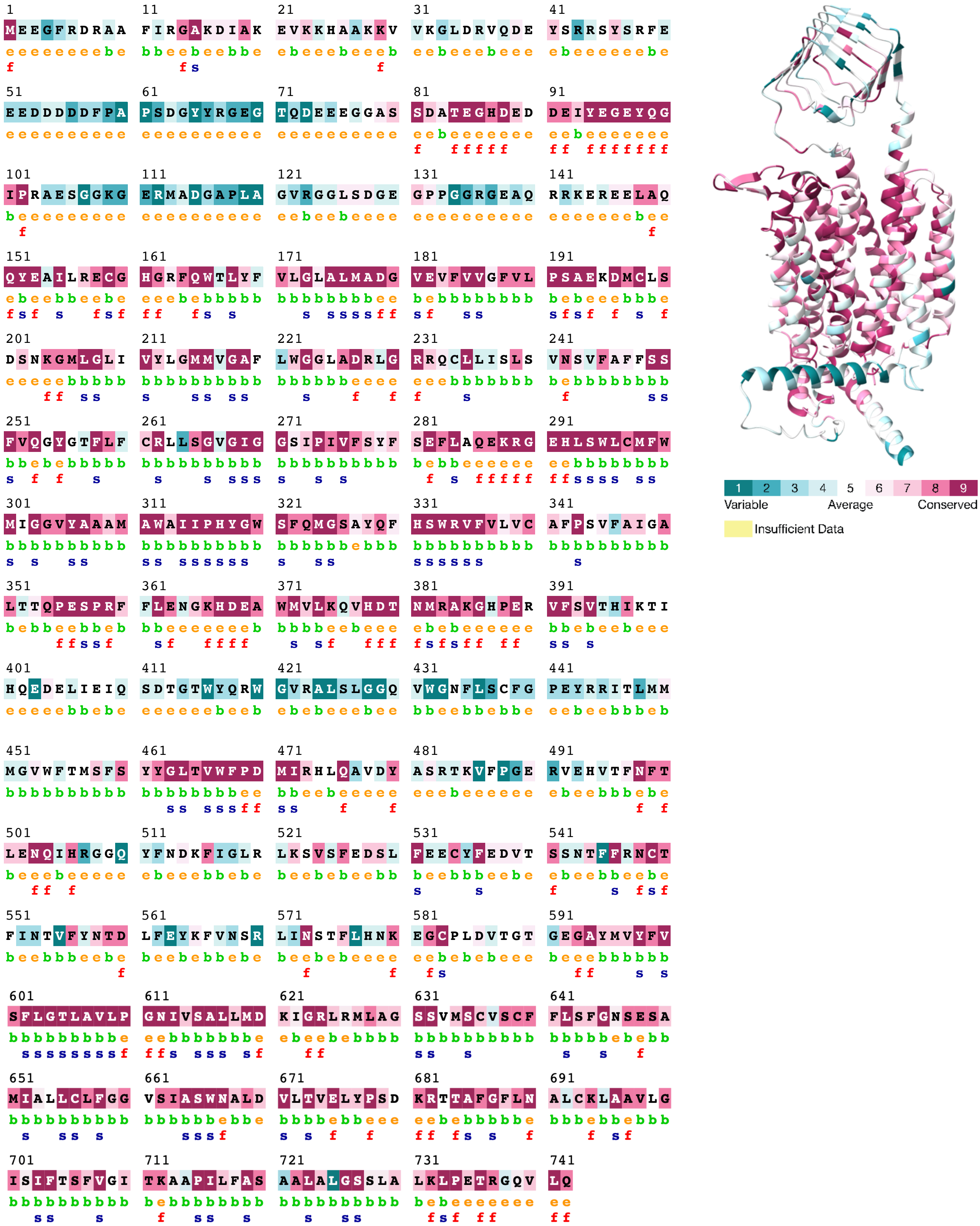


Figure S8. Synaptic vesicle glycoprotein 2A (SV2A) residue-wise evolutionary conservation profiles. Evolutionary conservation grades of each amino acid residue predicted by ConSurf server; visualized by the color-coding scheme of nine colors, ranging from turquoise (variable) through white (average) through burgundy (conserved) represents conservation grades 1 to 9, in order of increasing conservation (1= Variable, 5= Average, 9= Conserved). Conservation grades were calculated for the source amino acid sequence and the corresponding Alphafold3 predicted native structure. For clarity, the N- and C-termini and large loops were deleted.

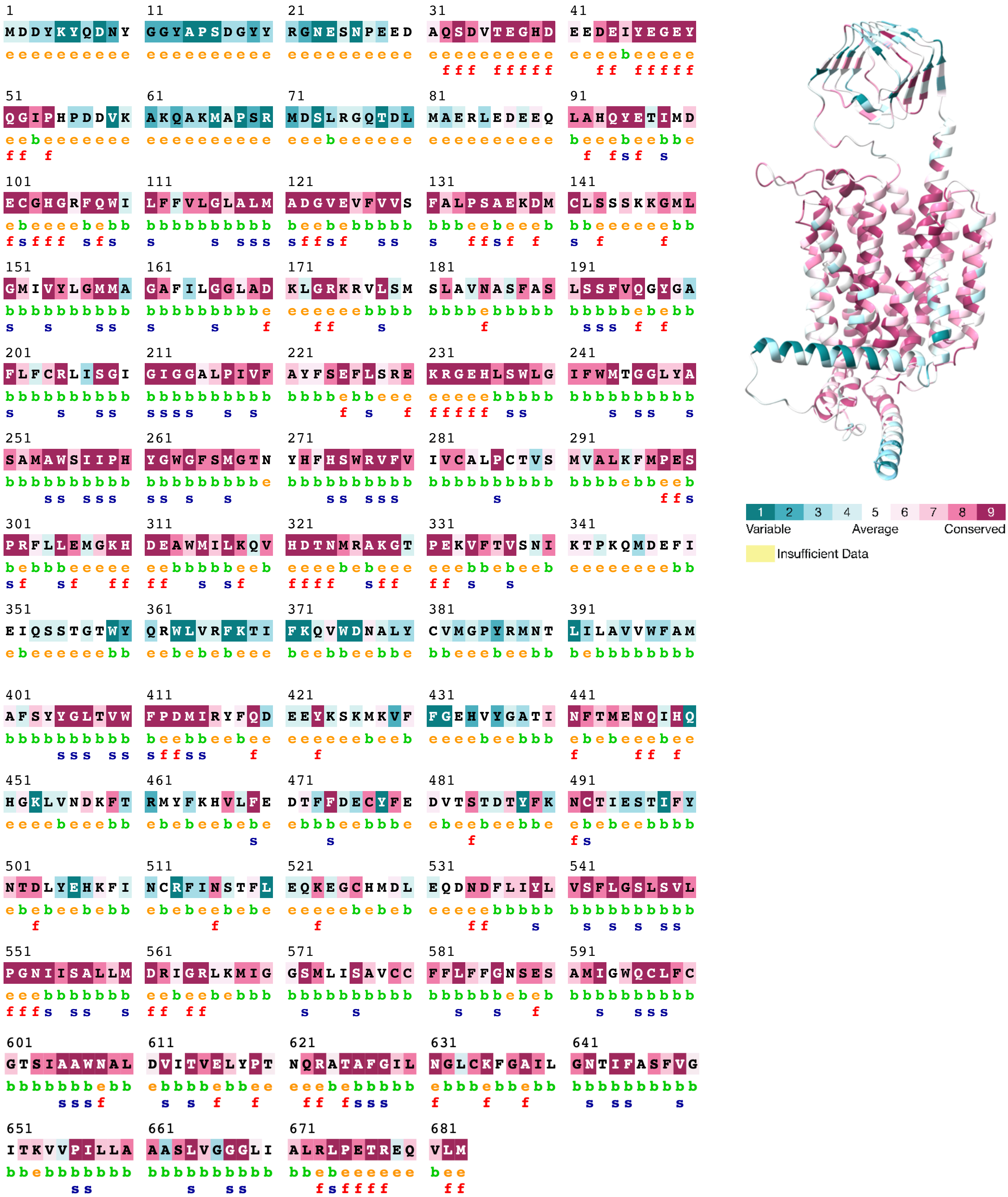


Figure S9. Synaptic vesicle glycoprotein 2B (SV2B) residue-wise evolutionary conservation profiles. Evolutionary conservation grades of each amino acid residue predicted by ConSurf server; visualized by the color-coding scheme of nine colors, ranging from turquoise (variable) through white (average) through burgundy (conserved) represents conservation grades 1 to 9, in order of increasing conservation (1= Variable, 5= Average, 9= Conserved). Conservation grades were calculated for the source amino acid sequence and the corresponding Alphafold3 predicted native structure. For clarity, the N- and C-termini and large loops were deleted.

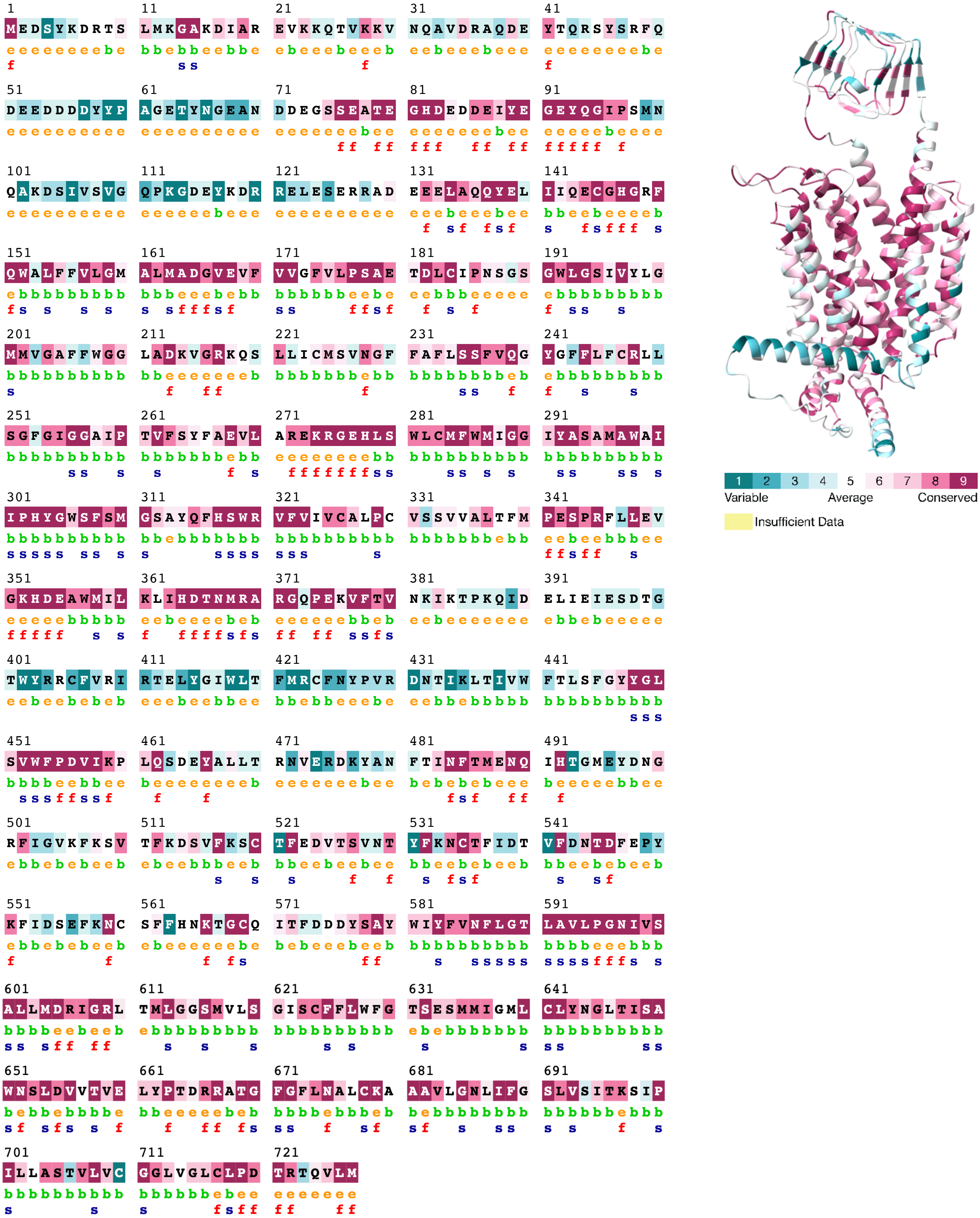


Figure S10. Synaptic vesicle glycoprotein 2C (SV2C) residue-wise evolutionary conservation profiles. Evolutionary conservation grades of each amino acid residue predicted by ConSurf server; visualized by the color-coding scheme of nine colors, ranging from turquoise (variable) through white (average) through burgundy (conserved) represents conservation grades 1 to 9, in order of increasing conservation (1= Variable, 5= Average, 9= Conserved). Conservation grades were calculated for the source amino acid sequence and the corresponding Alphafold3 predicted native structure. For clarity, the N- and C-termini and loops were deleted.

Figure S11. Synaptophysin (SYP) residue-wise evolutionary conservation profiles. Evolutionary conservation grades of each amino acid residue predicted by ConSurf server; visualized by the color-coding scheme of nine colors, ranging from turquoise (variable) through white (average) through burgundy (conserved) represents conservation grades 1 to 9, in order of increasing conservation (1= Variable, 5= Average, 9= Conserved). Conservation grades were calculated for the source amino acid sequence and the corresponding Alphafold3 predicted native structure. For clarity, the N- and C-termini and large loops were deleted.

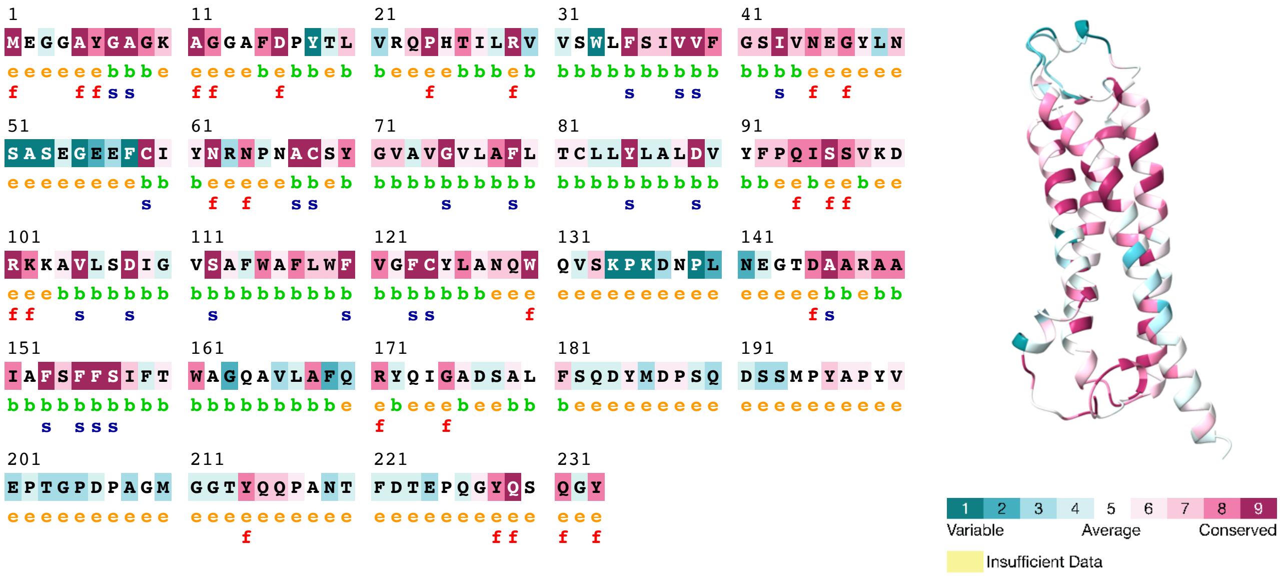


Figure S12. Synaptogyrin-1 (SYNGR1) residue-wise evolutionary conservation profiles. Evolutionary conservation grades of each amino acid residue predicted by ConSurf server; visualized by the color-coding scheme of nine colors, ranging from turquoise (variable) through white (average) through burgundy (conserved) represents conservation grades 1 to 9, in order of increasing conservation (1= Variable, 5= Average, 9= Conserved). Conservation grades were calculated for the source amino acid sequence and the corresponding Alphafold3 predicted native structure. For clarity, the N- and C-termini and large loops were deleted.

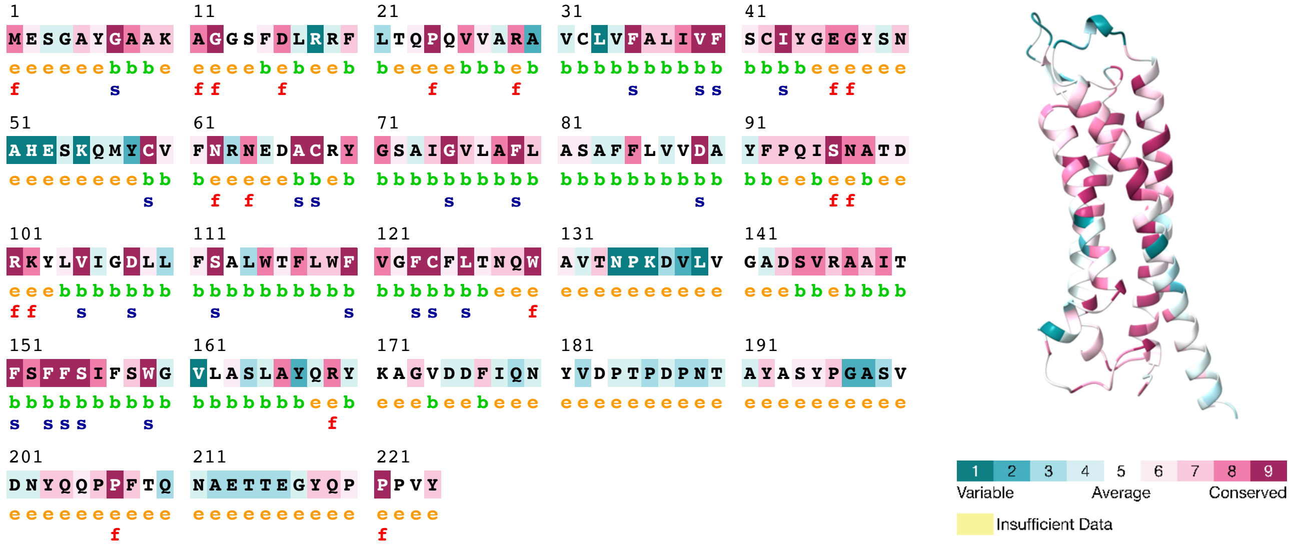


Figure S13. Synaptogyrin-2 (SYNGR2) residue-wise evolutionary conservation profiles. Evolutionary conservation grades of each amino acid residue predicted by ConSurf server; visualized by the color-coding scheme of nine colors, ranging from turquoise (variable) through white (average) through burgundy (conserved) represents conservation grades 1 to 9, in order of increasing conservation (1= Variable, 5= Average, 9= Conserved). Conservation grades were calculated for the source amino acid sequence and the corresponding Alphafold3 predicted native structure. For clarity, the N- and C-termini and large loops were deleted.

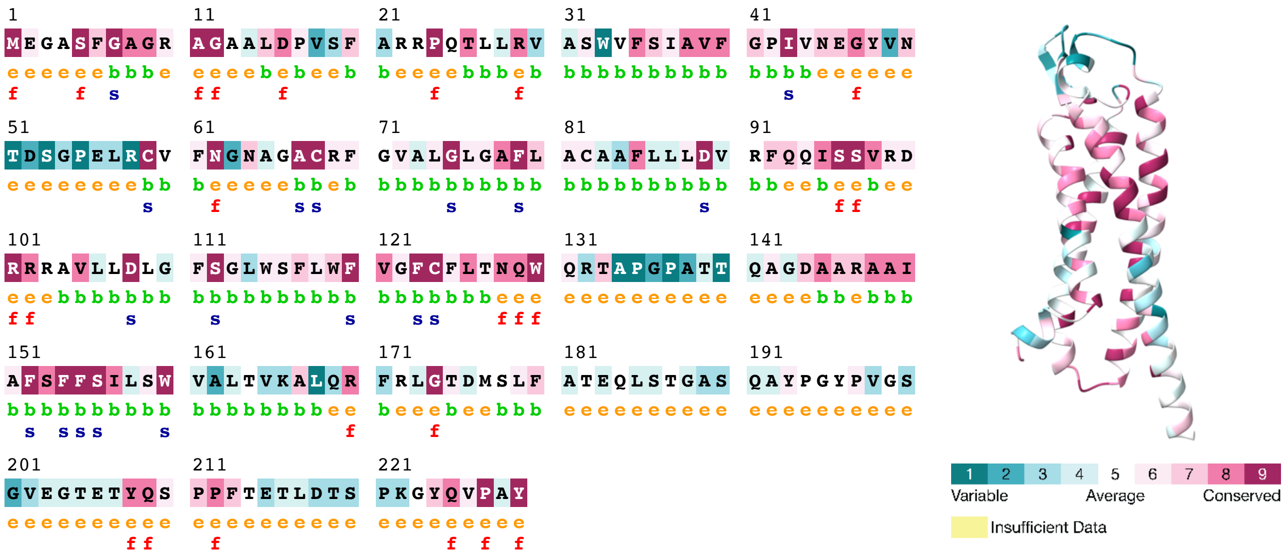


Figure S14. Synaptogyrin-3 (SYNGR3) residue-wise evolutionary conservation profiles. Evolutionary conservation grades of each amino acid residue predicted by ConSurf server; visualized by the color-coding scheme of nine colors, ranging from turquoise (variable) through white (average) through burgundy (conserved) represents conservation grades 1 to 9, in order of increasing conservation (1= Variable, 5= Average, 9= Conserved). Conservation grades were calculated for the source amino acid sequence and the corresponding Alphafold3 predicted native structure. For clarity, the N- and C-termini and large loops were deleted.

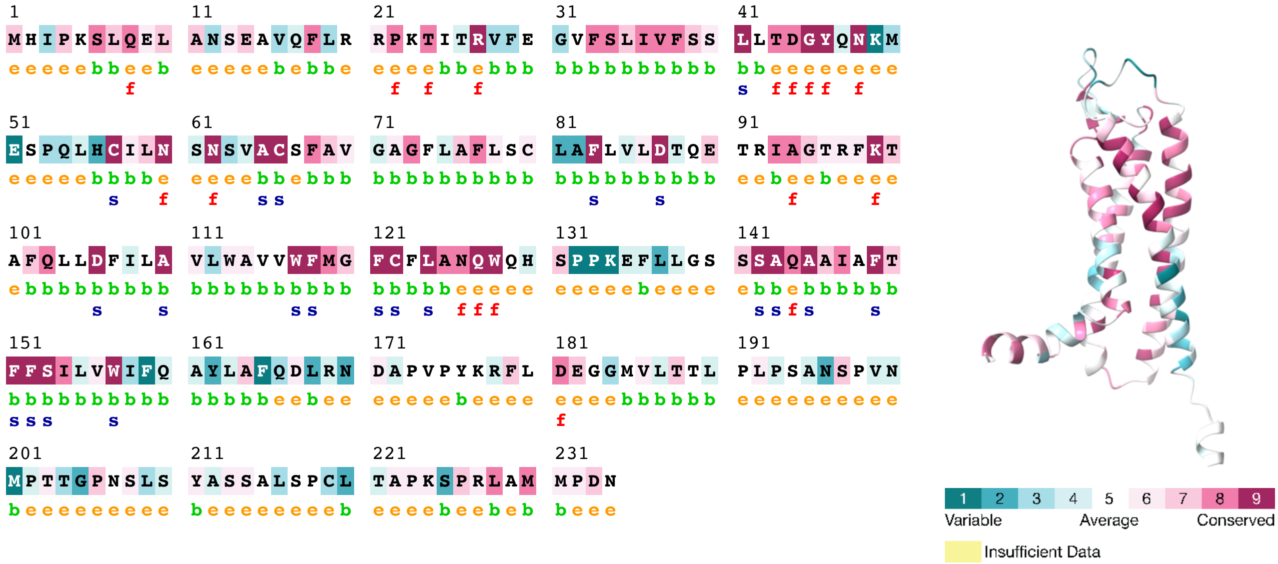


Figure S15. Synaptogyrin-4 (SYNGR4) residue-wise evolutionary conservation profiles. Evolutionary conservation grades of each amino acid residue predicted by ConSurf server; visualized by the color-coding scheme of nine colors, ranging from turquoise (variable) through white (average) through burgundy (conserved) represents conservation grades 1 to 9, in order of increasing conservation (1= Variable, 5= Average, 9= Conserved). Conservation grades were calculated for the source amino acid sequence and the corresponding Alphafold3 predicted native structure. For clarity, the N- and C-termini and large loops were deleted.

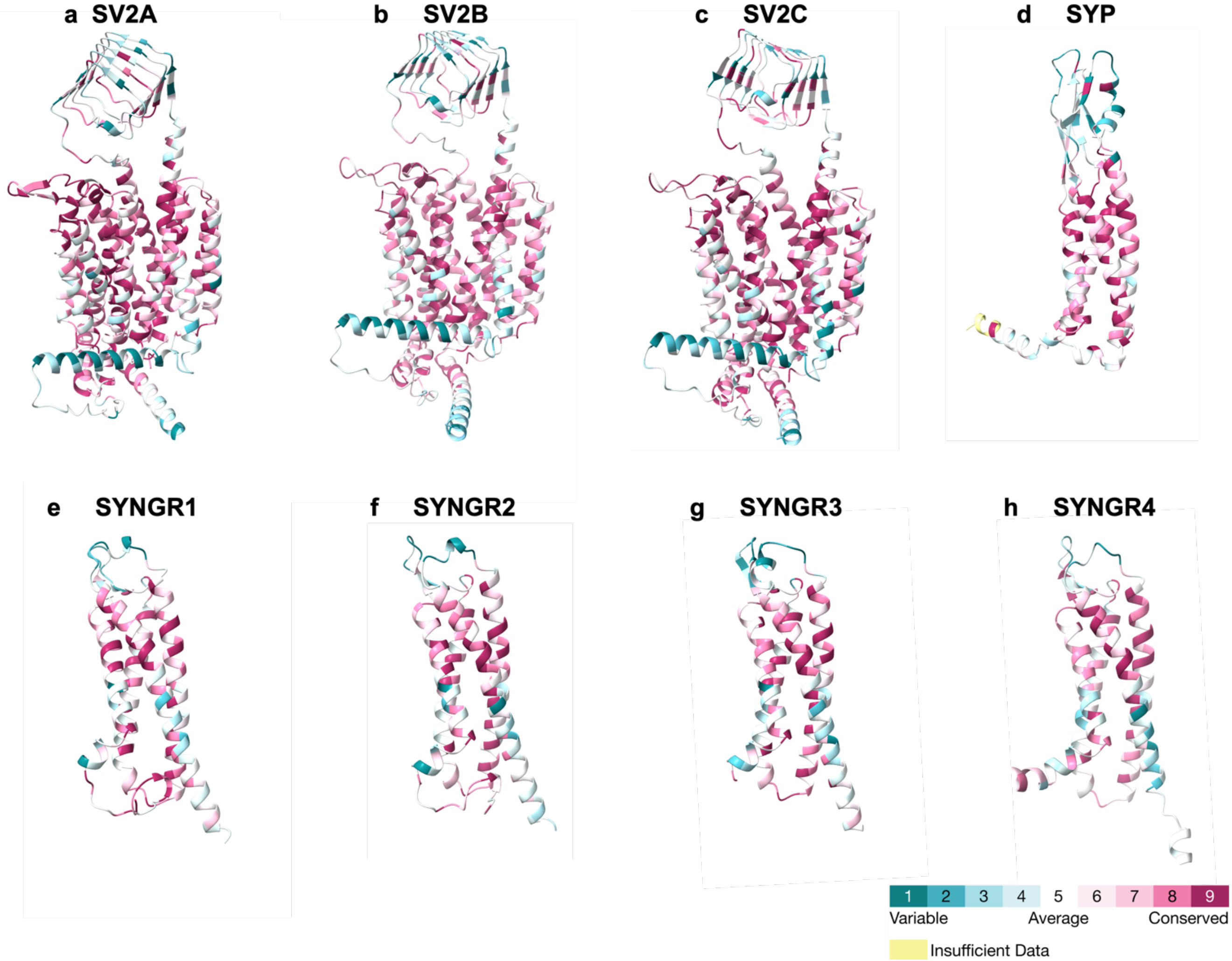
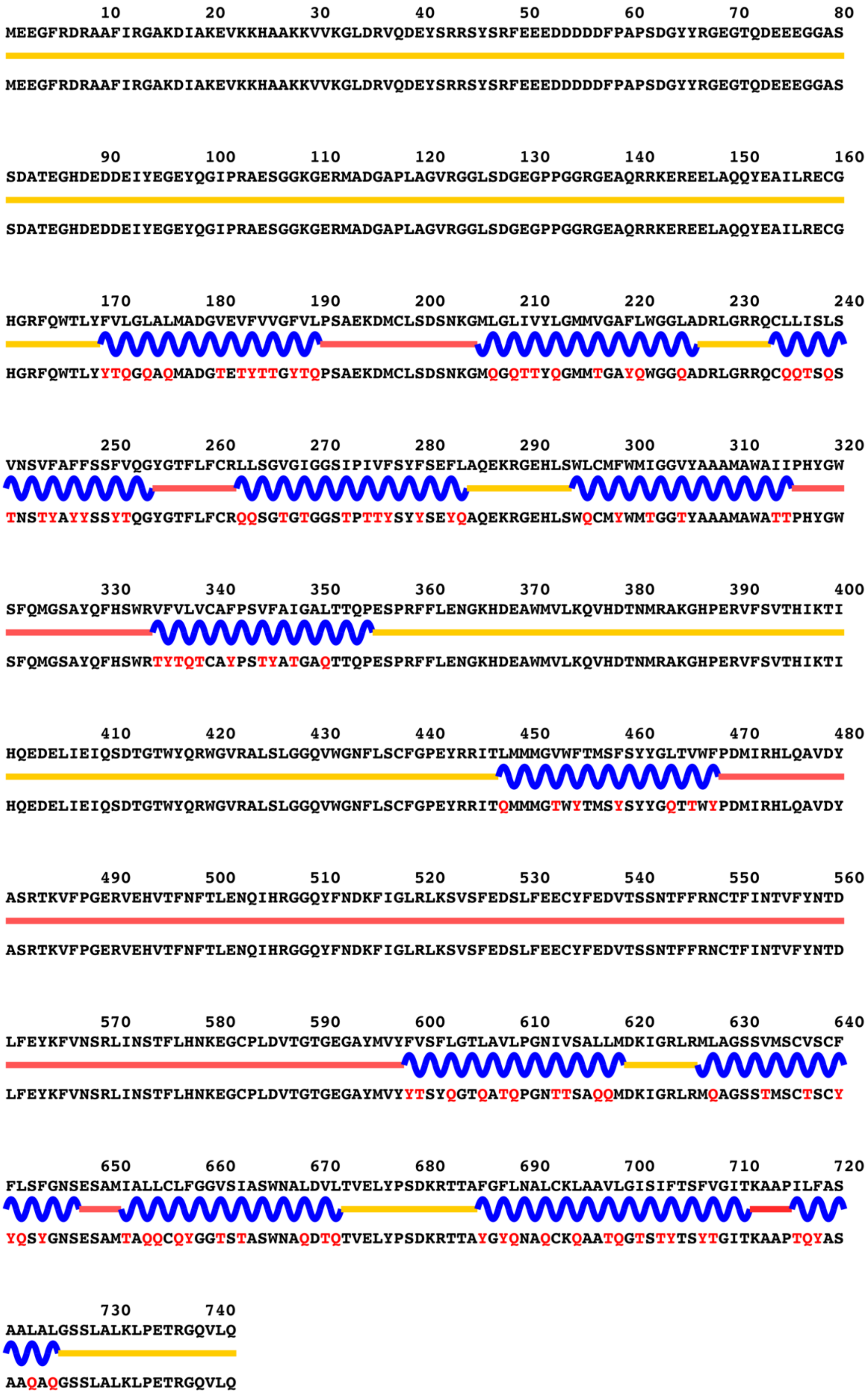
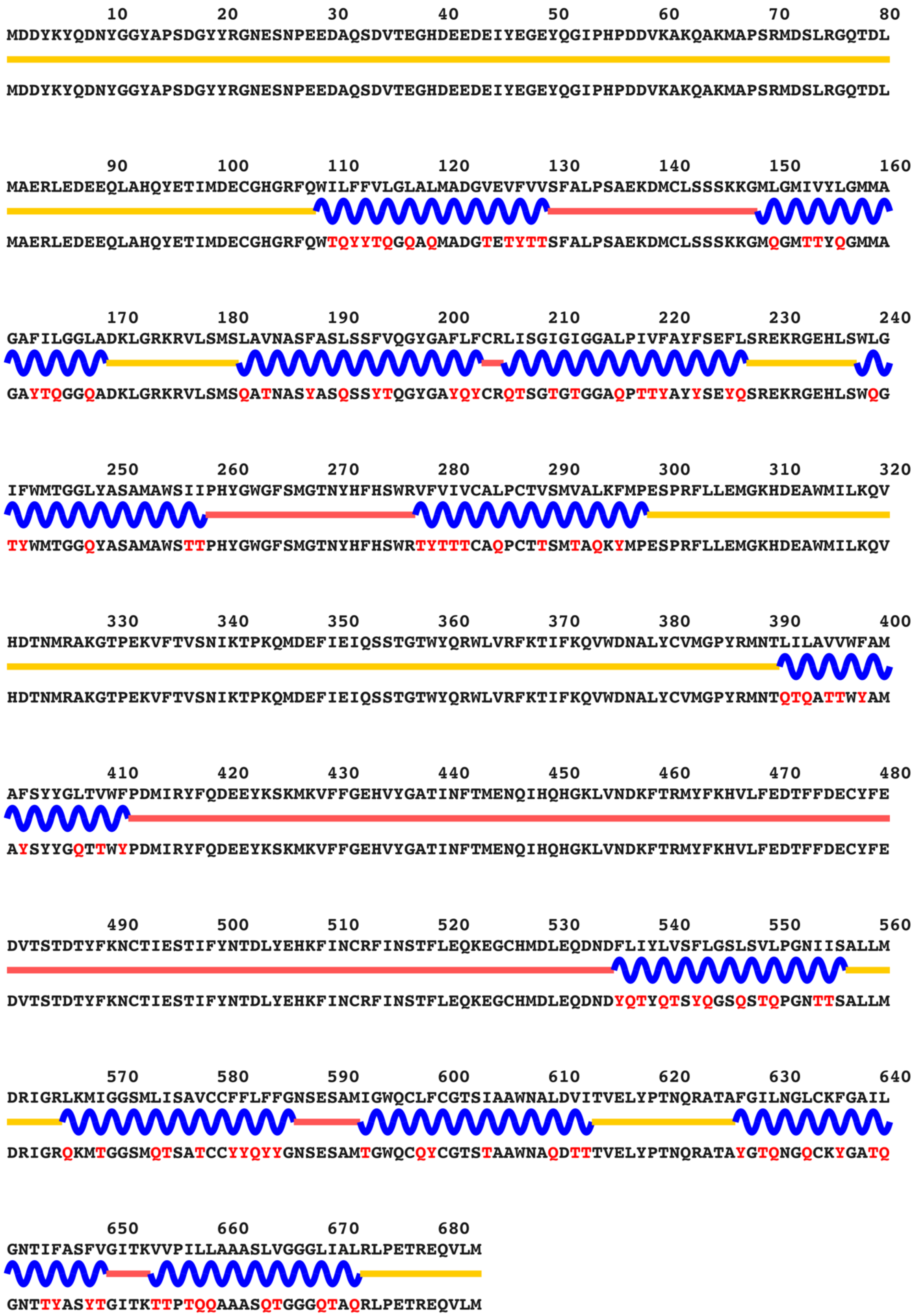


Figure S16. Evolutionary conservation profiles of 8 synaptic vesicle proteins including synaptic vesicle glycoprotein 2 family, synaptophysin, synaptogyrins. The predicted native structures and their residues colored by evolutionary conservation grades. The number of residues that have more than average conservation grade was calculated as follows: ~60.4% for SV2A (448/742), ~62.4% for SV2B (426/683), ~61.6% for SV2C (448/627), ~47.6% for SYP (149/313), ~54.5% for SYNGR1 (127/233), ~55.3% for SYNGR2 (124/224), ~54.1% for SYNGR3 (124/229), 50% for SYNGR4 (117/234). Conservation grades were calculated for the source amino acid sequence and the corresponding Alphafold3 predicted native structure. For clarity, the N- and C-termini and large loops were deleted in the figure.

Figure S17. Enlarged panel a-h of Figure 1. The QTY variant sequences are below the native protein sequences. The QTY amino acid substitution changes are colored in red. Other color code: Yellow line-intracellular, Blue wave-transmembrane helices, Pinkish line-extracellular.



a) SV2A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b) SV2B

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TRTQVLM

TRTQVLM

c) SV2C

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d) SYP

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e) SYNGR1

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f) SYNGR2

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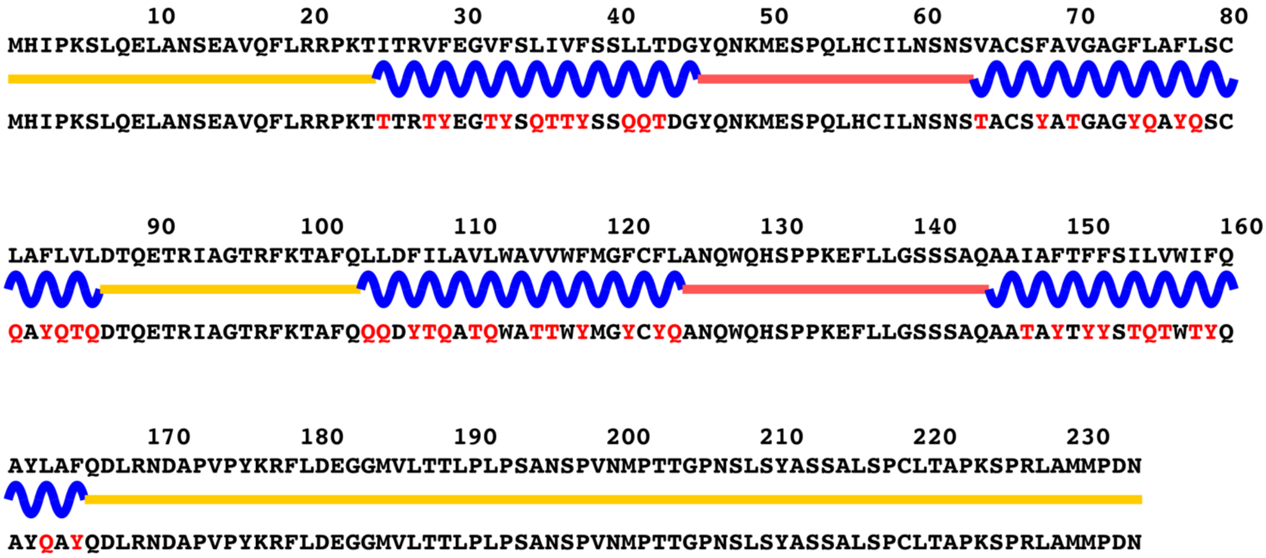
g) SYNGR3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h) SYNGR4